A WEAKLY ETALE RING MAP INDUCES SEPARABLE ALGEBRAIC RESIDUE FIELD EXTENSIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Weak etaleness is preserved by base change, so the fibre over a prime of the source is weakly etale over the residue field. invoked at proof lines 30561-30563. Reduces the statement to the case of a weakly etale algebra over a field.
- A weakly etale algebra over a field is a filtered colimit of etale algebras over that field. invoked at proof lines 30564-30566. Turns the weakly etale fibre into an object whose residue field extensions can be described.
- Etale algebras over a field are filtered colimits of finite separable field extensions, so their residue field extensions at primes are of that form. invoked at proof lines 30567-30571. Gives the conclusion that each residue field extension is a filtered colimit of finite separable extensions.

Source reference. Original lemma 092H: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30303–30303. Used at proof lines 30561-30563 to see that the fibre is weakly etale over the residue field; the reduction that follows is inside the counted span.

Source reference. Original lemma 092Q: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30504–30504. Invoked at proof lines 30564-30566 to write the fibre as a filtered colimit of etale algebras; the final passage to the residue field extension is inside the counted span.

Source reference. Original lemma 00U3: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 30567-30571 for the conclusion about the residue field extension; the identification of the extension as a filtered colimit of finite separable ones is made inside the counted span.

Lemma 0.2. *Let $A \to B$ be a ring map. If $A \to B$ is weakly étale, then $A \to B$ induces separable algebraic residue field extensions.*

Proof. Let $\mathfrak{p}$ be a prime of A. Then $\kappa(\mathfrak{p}) \to B \otimes_A \kappa(\mathfrak{p})$ is weakly étale by Lemma 092H. Hence $B \otimes_A \kappa(\mathfrak{p})$ is a filtered colimit of étale $\kappa(\mathfrak{p})$-algebras by Lemma 092Q. Hence for $\mathfrak{q} \subset B$ lying over $\mathfrak{p}$ the extension $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is a filtered colimit of finite separable extensions by Algebra, Lemma 00U3. □

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